



IILM UNIVERSITY
GREATER NOIDA, UTTAR PRADESH

UNSKILLCAMPUS VIRTUAL INTERNSHIP

Python Development

Weekly Progress Report Summary

Weeks 2 – 4 & Project Week 1 • Flask URL Shortener

REPORT SUBMISSION

3 August 2026

STUDENT CREDENTIALS

Abhay Raj

ENROLLMENT / ROLL NO.

2410030433

DEGREE

B.Tech (CSE)

ACADEMIC BATCH

2026 – 2027

HOST PLATFORM

UnskillCampus

Domain: Python Development

Candidate & Program Overview

Candidate Details

Student Name: Abhay Raj

Roll Number: 2410030433

Degree: B.Tech (Computer Science & Engineering)

Academic Batch: 2026–2027

University: IILM University, Greater Noida, U.P.

Internship Profile

Program Name:

UnskillCampus Virtual Internship

Specialization Domain:

Python Development

Current Coverage:

Week 2 – Week 4, Project Week 1

Report Submitted:

3 August 2026

Program Focus Areas

Python fundamentals • NumPy & Pandas • Skills assessment • Flask web development • SQLite integration

Internship Timeline at a Glance



Python Programming Fundamentals

Resource: Learning Python, 4th Edition — Mark Lutz | Video: if / if-else / if-elif-else

Tasks Completed

- Studied Python's syntax model and how the interpreter executes code
- Reviewed control-flow / conditional logic structure
- Learned if, if-else, and if-elif-else statement behavior
- Practiced small snippets: number ranges, grading logic, decision-making
- Reinforced Python's indentation-based block structure

Challenges & How They Were Solved

- Confusion between if-else vs if-elif-else → practiced varied examples
- Indentation errors → adopted consistent 4-space style
- Dense terminology (scoping, execution order) → re-read slowly, cross-referenced simpler sources

Key Lesson

Pairing a structured resource with a practical video walkthrough builds both conceptual depth and working coding ability.

Data Analysis using Python (NumPy & Pandas)

Focus: building a foundation in Python data-analysis libraries via guided video training

NumPy

- Array creation, indexing, slicing, reshaping
- Element-wise arithmetic & broadcasting
- Aggregation: sum, mean, max, min
- Matrix operations

Pandas

- Series & DataFrame structures
- Selection, filtering, sorting
- Grouping operations
- Handling missing values

Milestones

- Working understanding of NumPy array ops vs. Python lists
- Comfortable with basic–intermediate Pandas manipulation
- Local environment (Python, NumPy, Pandas) set up

Challenges: broadcasting shape rules, loc[] vs iloc[] indexing, and environment setup — resolved through re-watching training videos, small sample datasets, and official documentation.

Python Skills Assessment

A timed quiz evaluating core Python concepts covered so far.

Challenges

- Some questions were time-consuming and required careful reading
- A couple of questions touched on concepts not recently revised

Lessons Learned

- Skim all questions first to prioritize easier ones under time pressure
- Regular revision of core topics builds confidence in timed tests
- Reviewing mistakes after an assessment reinforces learning

SCORE ACHIEVED

18 / 20

10 Questions • 30 Minutes

Plan for Next Week

- Revise topics where marks were lost
- Continue problem-solving practice
- Prepare for upcoming project tasks

URL Shortener — Flask Web Application

Building and structuring a Flask app with SQLite-backed URL shortening and redirection

1 Project Setup & Structure

Organized app.py, templates/index.html, database.db, and requirements.txt; configured the Flask development server.

2 Database Integration

Built init_db() with sqlite3 to auto-create the urls table (id, original, short) on startup.

3 Short URL Generation & Redirection

generate_code() creates 6-character alphanumeric codes; "/" handles form display and submission; "/<code>" redirects or returns "URL Not Found".

4 Front-End Interface

Styled HTML page with a submission form and a clickable generated short-URL display.

Challenges & Next Steps

Current Challenges

- Database connection handling: opening/closing SQLite connections per request adds repetitive code
- Short code reliability: generation logic doesn't check for collisions, risking duplicate codes

Learning Resources Used

Official Flask documentation (routing, requests, templates) and Python standard library docs for sqlite3, random, and string.

Next Week's Goals

- Add URL validation and graceful duplicate handling
- Add uniqueness check / retry mechanism for short codes
- Introduce click-tracking and analytics per short URL
- Support custom short codes
- Improve front-end styling and user experience

Key Skills Demonstrated

1

Core Python

Control flow, conditionals, syntax fundamentals

2

Data Analysis

NumPy arrays & Pandas DataFrames for tabular data

3

Self-Assessment

Scored 18/20 on a timed Python concepts quiz

4

Web Development

Flask routing, templates, and app structuring

5

Database Integration

SQLite schema design and query handling

6

Debugging & Docs

Systematic troubleshooting using official documentation

Thank You!

Abhay Raj

Python Development Intern

UnskillCampus

Internship Platform